

RQ Answers

omitted

2024-04-05

Data overview

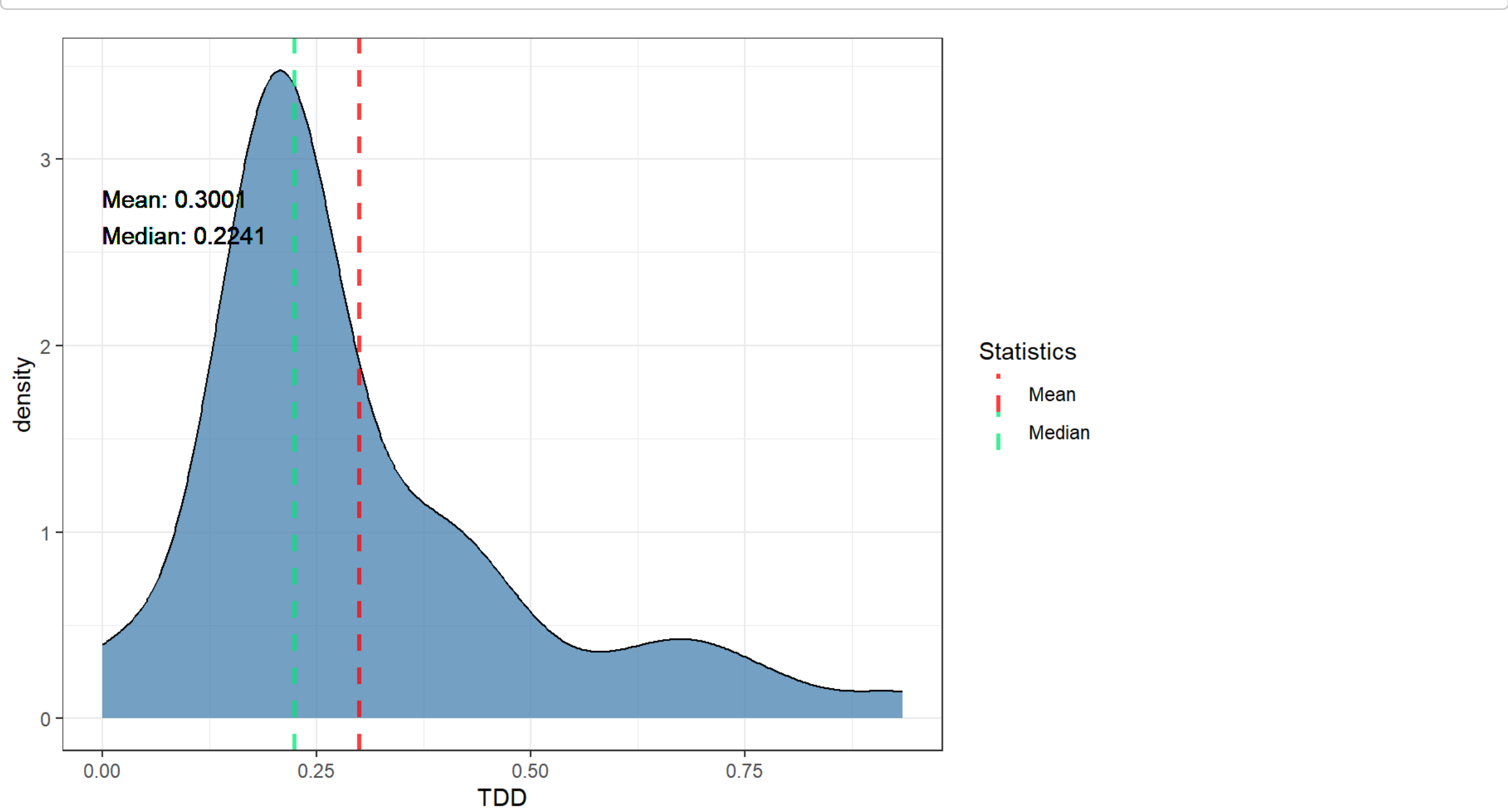
glimpse(all_data)				
## Rows: 957				
## Columns: 11				
## \$ repo	<chr>	"commons-io", "commons-io", "commons-io", "commons...		
## \$ pr_number	<int>	177, 177, 177, 177, 177, 177, 177, 177, 177, ...		
## \$ rule	<chr>	"java:S1104", "java:S1444", "java:S3008", "java:S1...		
## \$ severity	<chr>	"MINOR", "MINOR", "MINOR", "CRITICAL", "INFO", "CR...		
## \$ file	<chr>	"src/main/java/org/apache/commons/io/file/NoopPath...		
## \$ type	<chr>	"CODE_SMELL", "CODE_SMELL", "CODE_SMELL", "CODE_SM...		
## \$ status	<chr>	"CLOSED", "CLOSED", "CLOSED", "OPEN", "OPEN", "OPE...		
## \$ debt	<int>	10, 20, 2, 30, 0, 30, 8, 20, 5, 14, 5, 28, 10, 5, ...		
## \$ ncloc_affected_file	<int>	4, 4, 4, 112, 38, 100, 100, 1122, 1122, 1122, 1122...		
## \$ complexity	<int>	0, 0, 0, 28, 6, 25, 25, 301, 301, 301, 301, 3...		
## \$ origin	<chr>	"PRE-EXISTING", "PRE-EXISTING", "PRE-EXISTING", "P...		

RQ1

TDD (Technical Debt Density) by PR Distribution

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Mean:", round(mean(tdd), : All aesthetics have length 1, but the data has 50 rows.
## | Please consider using `annotate()` or provide this layer with data containing
##   a single row.
```

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Median:", round(median(tdd), : All aesthetics have length 1, but the data has 50 rows.
## | Please consider using `annotate()` or provide this layer with data containing
##   a single row.
```

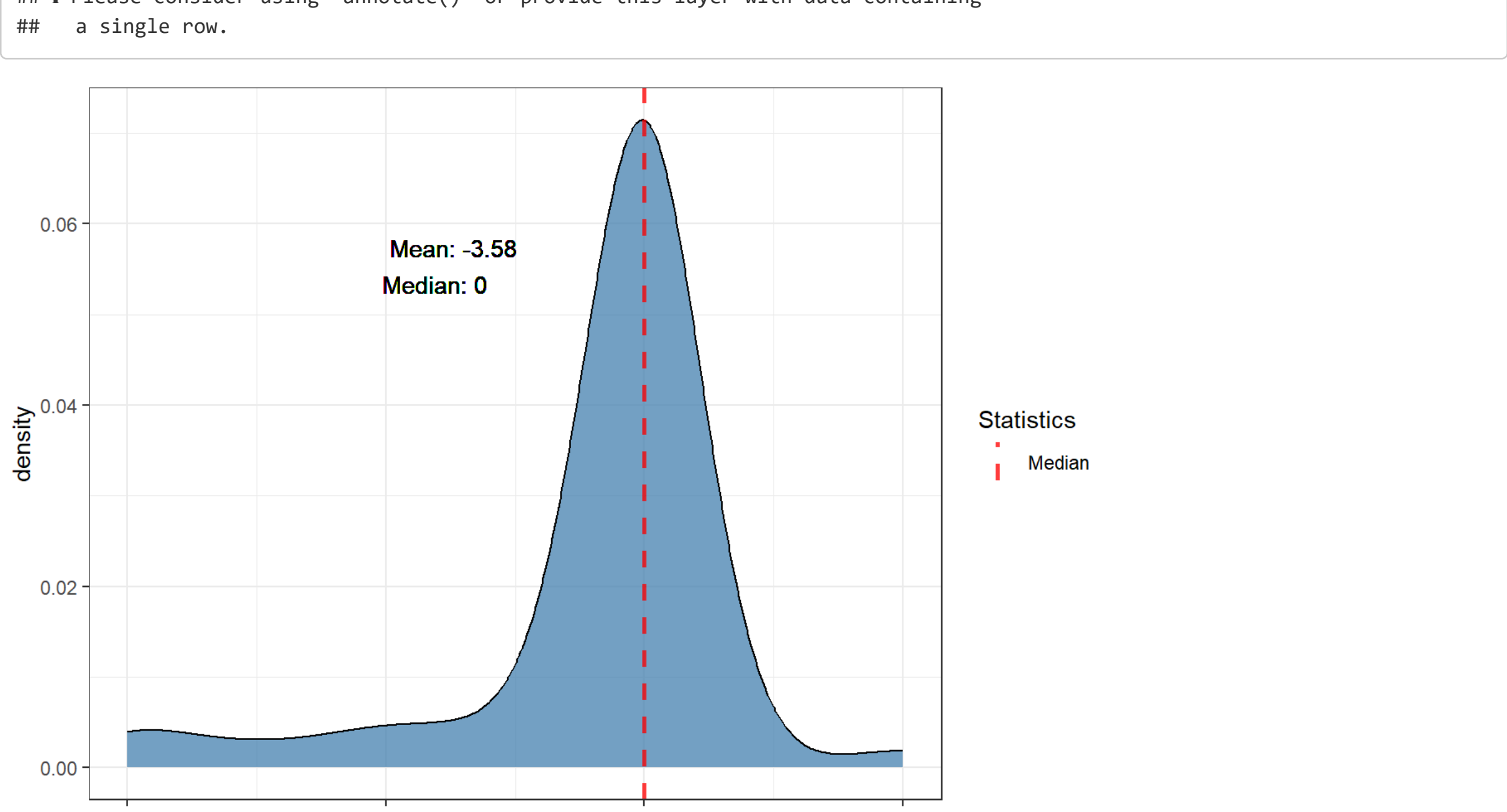


TDV (Technical Debt Variation)

TDV Distribution

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Mean:", round(mean(tdv), : All aesthetics have length 1, but the data has 50 rows.
## | Please consider using `annotate()` or provide this layer with data containing
##   a single row.
```

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Median:", round(median(tdv), : All aesthetics have length 1, but the data has 50 rows.
## | Please consider using `annotate()` or provide this layer with data containing
##   a single row.
```



Percentages for TDV < 0 (TD decrease), TDV = 0 (TD unchanged) and TDV > 0 (TD increased) by repo

```
## # A tibble: 1 x 4
##   repo      tdv_lower_than_zero tdv_equal_to_zero tdv_greater_than_zero
##   <chr>                <dbl>          <dbl>          <dbl>
## 1 commons-io                22             72             6
```

Mean of percentages of TDV (mean of repo percentages)

```
## # A tibble: 1 x 3
##   mean_tdv_lower_than_zero mean_tdv_equal_to_zero mean_tdv_greater_than_zero
##   <dbl>                <dbl>          <dbl>          <dbl>
## 1                22             72             6
```

Percentages of pre-existing TDV by repo

```
## # A tibble: 1 x 3
##   repo      preexisting_tdv_equal_to_zero preexisting_tdv_greater_than_zero
##   <chr>                <dbl>          <dbl>          <dbl>
## 1 commons-io                76             24
```

Mean of percentages of pre-existing TDV (mean of repo percentages)

```
## # A tibble: 1 x 2
##   preexisting_tdv_equal_to_zero mean_preexisting_tdv_greater_than_zero
##   <dbl>                <dbl>          <dbl>          <dbl>
## 1                76             24
```

Percentages for new TDV by repo

```
## # A tibble: 1 x 3
##   Groups:   repo [1]
##   repo      new_tdv_equal_to_zero new_tdv_greater_than_zero
##   <chr>                <dbl>          <dbl>          <dbl>
## 1 commons-io                77.8             22.2
```

Mean of percentages of new TDV (mean of repo percentages)

```
## # A tibble: 1 x 2
##   mean_new_tdv_equal_to_zero mean_new_tdv_greater_than_zero
##   <dbl>                <dbl>          <dbl>          <dbl>
## 1                77.8             22.2
```

RQ2

As data is unbalanced across repositories, as shown in the issue characterization (`./issues_characterization.Rmd`), we will examine the top 10 fixed issues and the top 10 unfixed issues for each repository, and then aggregate them into a rank using the *PositionScore* metric given by the following formula:

$$PositionScore_i = \sum_{j=1}^k Position_{ij}$$

given a rule i , its *PositionScore_i* will be the sum of its position across all k repositories. After calculating the metrics for all rules, we rank them in a TOP 10 of the most frequent rules (fixed or unfixed, depending on the context) across repositories. The lower the PositionScore, the more frequent the rule across repositories.

TOP 10 fixed per repo ranked by count

All top 10 fixed rules for each repo.

```
## # A tibble: 10 x 2
##   Groups:   rank [10]
##   rank `commons-io`
##   <int> <chr>
## 1     1 java:S1125 - 8
## 2     2 java:S1612 - 4
## 3     3 java:S1192 - 4
## 4     4 java:S1135 - 3
## 5     5 java:S1128 - 2
## 6     6 java:S1066 - 2
## 7     7 java:S3008 - 2
## 8     8 java:S1191 - 1
## 9     9 java:S1130 - 1
## 10    10 java:S3551 - 1
```

TOP 10 fixed

TOP 10 fixed rules by PositionScore metric.

```
## # A tibble: 10 x 2
##   rule      position_score
##   <chr>                <dbl>
## 1 java:S1125            122
## 2 java:S1192            123
## 3 java:S1612            124
## 4 java:S1135            125
## 5 java:S1066            126
## 6 java:S1128            127
## 7 java:S3008            128
## 8 java:S1104            129
## 9 java:S1130            130
## 10 java:S1133            131
```

TOP 10 unfixed per repo ranked by count

All top 10 unfixed rules for each repo.

```
## # A tibble: 10 x 2
##   Groups:   rank [10]
##   rank `commons-io`
##   <int> <chr>
## 1     1 java:S1133 - 255
## 2     2 java:S1192 - 136
## 3     3 java:S3776 - 136
## 4     4 java:S127 - 72
## 5     5 java:S1135 - 65
## 6     6 java:S1130 - 50
## 7     7 java:S4042 - 39
## 8     8 java:S2183 - 38
## 9     9 java:S1125 - 25
## 10    10 java:S3626 - 20
```

TOP 10 unfixed

TOP 10 unfixed rules by PositionScore metric.

```
## # A tibble: 10 x 2
##   rule      position_score
##   <chr>                <dbl>
## 1 java:S1133            122
## 2 java:S1192            123
## 3 java:S3776            124
## 4 java:S127            125
## 5 java:S1135            126
## 6 java:S1130            127
## 7 java:S4042            128
## 8 java:S2183            129
## 9 java:S1125            130
## 10 java:S3626            131
```

Outliers

Higher TDD

```
## # A tibble: 50 x 5
##   repo      pr_number total_debt total_ncloc  tdd
##   <chr>        <int>      <int>      <int> <dbl>
## 1 commons-io      587         85         91 0.934
## 2 commons-io     205        115        144 0.799
## 3 commons-io     530        120        168 0.714
## 4 commons-io     452         30         43 0.698
## 5 commons-io     566        165        257 0.642
## 6 commons-io     325         30         50 0.6
## 7 commons-io     521         90        177 0.508
## 8 commons-io      19        258        533 0.484
## 9 commons-io     453         30         68 0.441
## 10 commons-io      79        282        651 0.433
## # i 40 more rows
```

Higher TDV

```
## # A tibble: 50 x 3
##   pr_number repo      tdv
##   <int> <chr>      <int>
## 1     194 commons-io      20
## 2     529 commons-io       7
## 3     558 commons-io       1
## 4      19 commons-io       0
## 5      43 commons-io       0
## 6      79 commons-io       0
## 7     201 commons-io       0
## 8     236 commons-io       0
## 9     245 commons-io       0
## 10    267 commons-io       0
## # i 40 more rows
```

Lower TDV

```
## # A tibble: 50 x 3
##   pr_number repo      tdv
##   <int> <chr>      <int>
## 1     205 commons-io     -40
## 2     233 commons-io     -39
## 3     177 commons-io     -32
## 4     491 commons-io     -25
## 5     530 commons-io     -20
## 6     243 commons-io     -12
## 7     582 commons-io      -8
## 8     179 commons-io      -5
## 9     566 commons-io      -5
## 10    566 commons-io      -5
## # i 40 more rows
```

Extra

Number of PRs with pre-existing TD

```
##   n
## 1 50
```

Number of java:S1192 issues that affect test code

```
##   n
## 1 10
```

Number of java:S1192 issues that affect not test code

```
##   n
## 1 140
```

Summary NCLOC

```
##   ncloc
##   Min.   : 43.0
##   1st Qu.: 187.5
##   Median : 657.5
##   Mean   : 765.4
##   3rd Qu.: 1017.0
##   Max.   : 2787.0
```